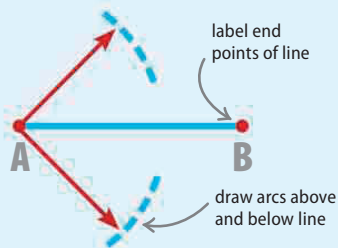
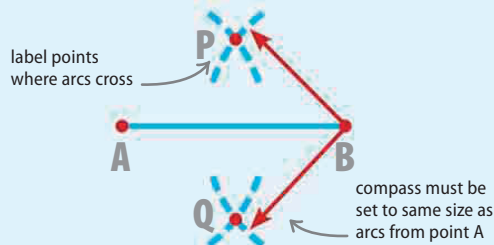


Constructing 90° and 45° angles

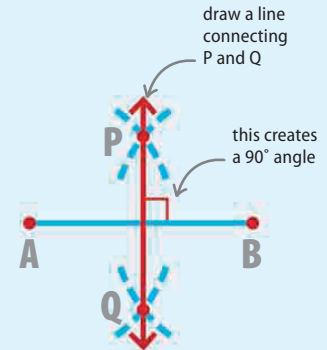
Bisecting an angle can be used to construct some common angles without using a protractor, for example a right angle (90°) and a 45° angle.



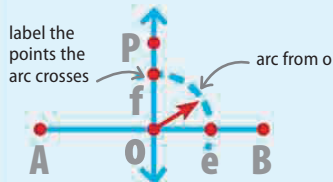
Draw a straight line (AB). Place a compass on point A, set it to a distance just over half of the line's length, and draw an arc above and below the line.



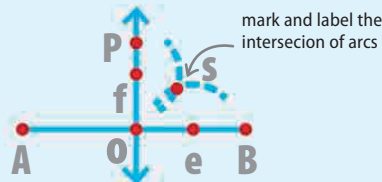
Then, draw two arcs with the compass set to the same length and placed on point B. Label the points where the arcs cross each other P and Q.



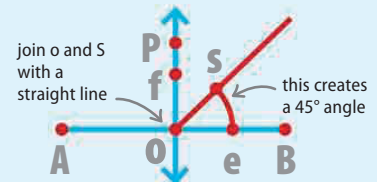
Draw a line from point P to point Q. This is a perpendicular bisector of the original line and it creates four 90° angles.



Draw an arc from point o that crosses two lines on either side, this creates a 45° angle. Label the two points where the arc intersects the lines, f and e.



Keep the compass at the same length as the last arc and draw arcs from points f and e. Label the intersection of these arcs with a letter (s).



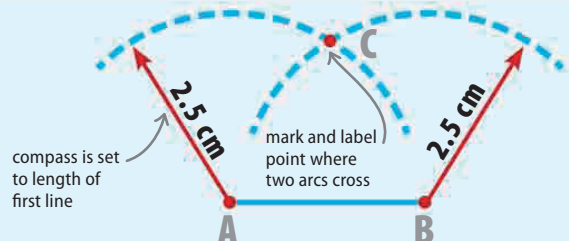
Draw a line from point o through s. This line is the angle bisector. The 90° angle is now split into two 45° angles.

Constructing 60° angles

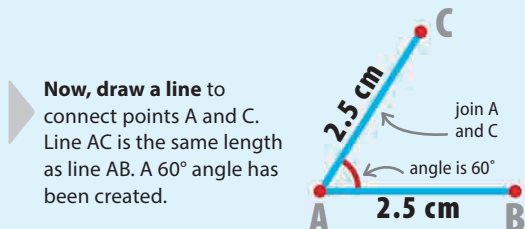
An equilateral triangle, which has three equal sides and three 60° angles, can be constructed without a protractor.



Draw a line, which will form one arm of the first angle. Here the line is 2.5 cm long, but it can be any length. Mark each end of the line with a letter.



Now, set the compass to the same length as the first line. Draw an arc from point A, then another from point B. Mark the point where the two arcs cross, C.



Now, draw a line to connect points A and C. Line AC is the same length as line AB. A 60° angle has been created.

Construct an equilateral triangle by drawing a third line from B to C. Each side of the triangle is equal and each internal angle of the triangle is 60°.

